Network Design Specification Document

Group 5

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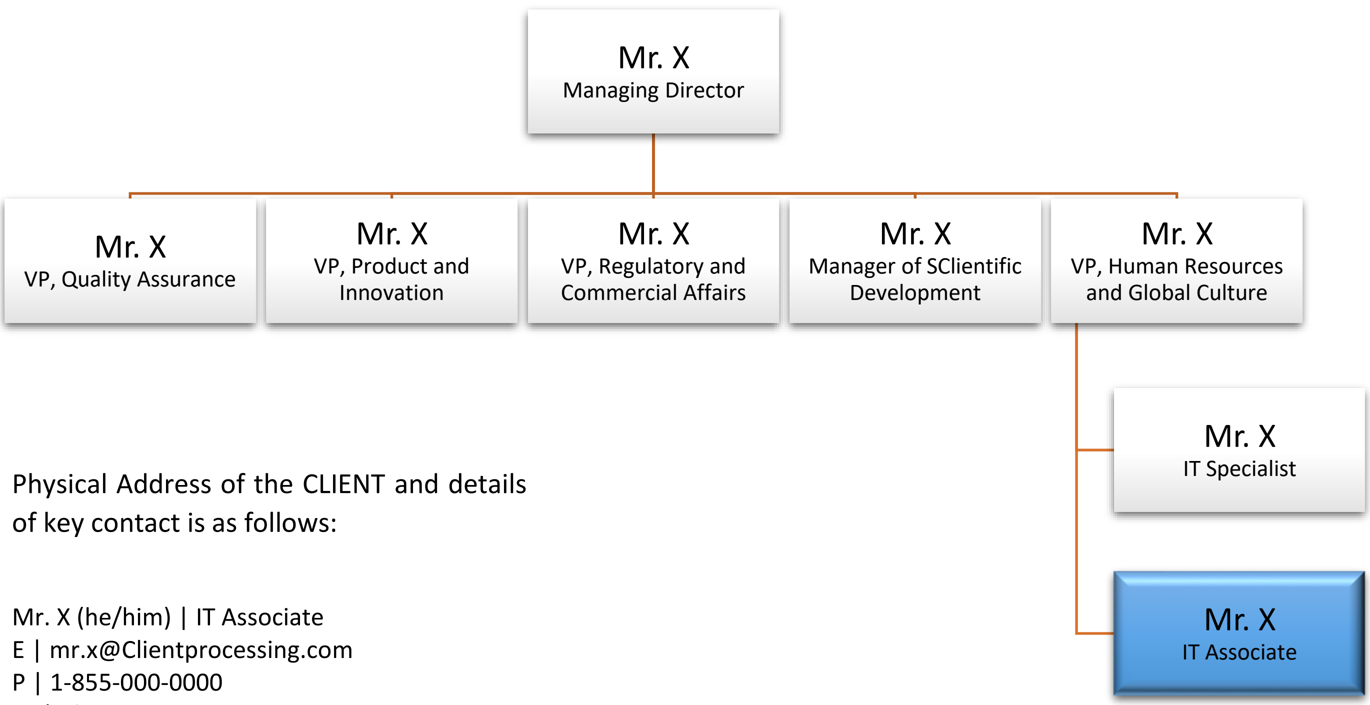
Executive Overview:

CLIENT is a company that offers comprehensive cannabis product solutions for both global brand leaders and startups seeking an innovative edge. With a focus on specialist formulations and product development, as well as extensive experience working with regulators and wholesalers, CLIENT is deeply embedded in the fast-growing legal cannabis market.

Leveraging years of cannabis experience, CLIENT has the necessary capabilities and connections to help clients analyze, validate, and capitalize on new opportunities in the industry. Regardless of the stage of a brand's journey, clients can tap into CLIENT's expertise to increase their chances of success.

Overall, CLIENT is positioned to help clients establish the next high-demand brand in the legal cannabis industry, providing the key to unlocking every possibility.

The key contact from within the **CLIENT**, who is directly involved in this project is *Mr. X (IT Associate)* is highlighted in the Organization Chart below.



Physical Address of the CLIENT and details of key contact is as follows:

Mr. X (he/him) | IT Associate
E | mr.x@Clientprocessing.com
P | 1-855-000-0000
W | Clientprocessing.com
Oldcastle, ON N0R 1L0

Figure 1

Executive Overview (continued):

The ensuing record signifies the fulfillment of CLIENT's Network Design process. By thoroughly examining the current Network infrastructure and its functionalities, along with a Traffic Analysis, PTHK has provided Client with significant advice and remedies. These recommendations and solutions will enable the required modifications to enhance network effectiveness and reduce the Total Cost of Ownership.

The following Diagram represents the 5 Phase Network Design Process.

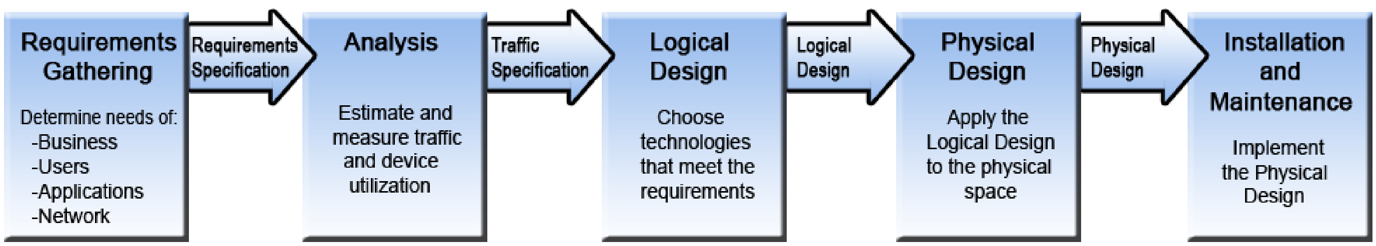


Figure 2

The **Requirements Gathering Phase**, which involves collecting and recording data on the network's needs and goals, is the most crucial stage of the network design process. To precisely define the needs of CLIENT, this step entails collecting data from stakeholders, end users, and other team members. With the use of this data, a thorough network design strategy that considers the requirements of both stakeholders and end users will be developed. It is possible to establish what consumers want and need during the requirements gathering phase, which must consider both the organization's present and future demands.

The Requirements Gathering phase provides the below results:

- Better understanding of the current network.
- Facilitate objective decision-making.
- Enabling the planning of network migration.
- Delivering suitable resources to all users.

The **Traffic Analysis Phase** of the Network Design Process involves collecting and analyzing data about the network's traffic to identify performance issues, security risks, or scalability challenges. To gather data and examine network traffic, it makes use of specialized instruments like packet analyzers and network monitoring software. The deliverables of the traffic analysis phase include a thorough report outlining the present and foreseeable traffic patterns and network requirements, including details on the traffic type, volume, protocols, and technologies to be used, as well as the hardware and software required. Additionally, it offers suggestions for how to configure the network to best meet the demands.

The Traffic Analysis Phase provides the following information:

- Identification of network traffic patterns.
- Identification of performance bottlenecks, congestion points or other issues.
- Development of a comprehensive network design plan.
- Implementation of changes to network's topology.

Executive Overview (continued):

The **Logical Design Phase** involves specifying the network topology, protocols, and addressing schemes to ensure the network complies with the requirements acquired in prior phases. The technical needs of the network, such as the anticipated availability, traffic type, and user count, must be understood at this phase. The outputs from the logical design phase frequently include a thorough design document, diagrams, flowcharts, and other visual aids that illustrate how the network will be configured. The document should include providing a budget for the project, as well as an implementation schedule and strategy.

The Logical Design Phase provides the below results:

- Development of a high-level conceptual view of the network.
- Definition of network topology, protocols and addressing schemes.
- Creation of a flexible and scalable network framework.

The **Physical Design Phase** of network design entails developing a thorough strategy for the execution of the network, which includes choosing hardware and software components and figuring out where and how they should be placed. It is the step when the high-level logical design is converted into a thorough physical design plan that outlines the physical elements required to accomplish the logical design.

The Physical Design Phase gives the below information:

- Selection of specific hardware and software components.
- Determination of their placement and configuration within network.
- Consideration of physical constraints.
- Detailed documentation of network topology.
- Definition of procedures for installation.

The **Installation and Maintenance Phase**, the last stage of the network design process involves physically deploying network components and managing and supporting the network infrastructure. It includes testing and maintaining network as well as its troubleshooting.

The Installation and Maintenance provides the below information:

- Deployment of physical network components.
- Configuration and testing of network.
- Ensuring network performance, security, and availability.
- Implementation of troubleshooting, upgrades, and backup procedures.
- Ensuring the network meets the organization’s needs.

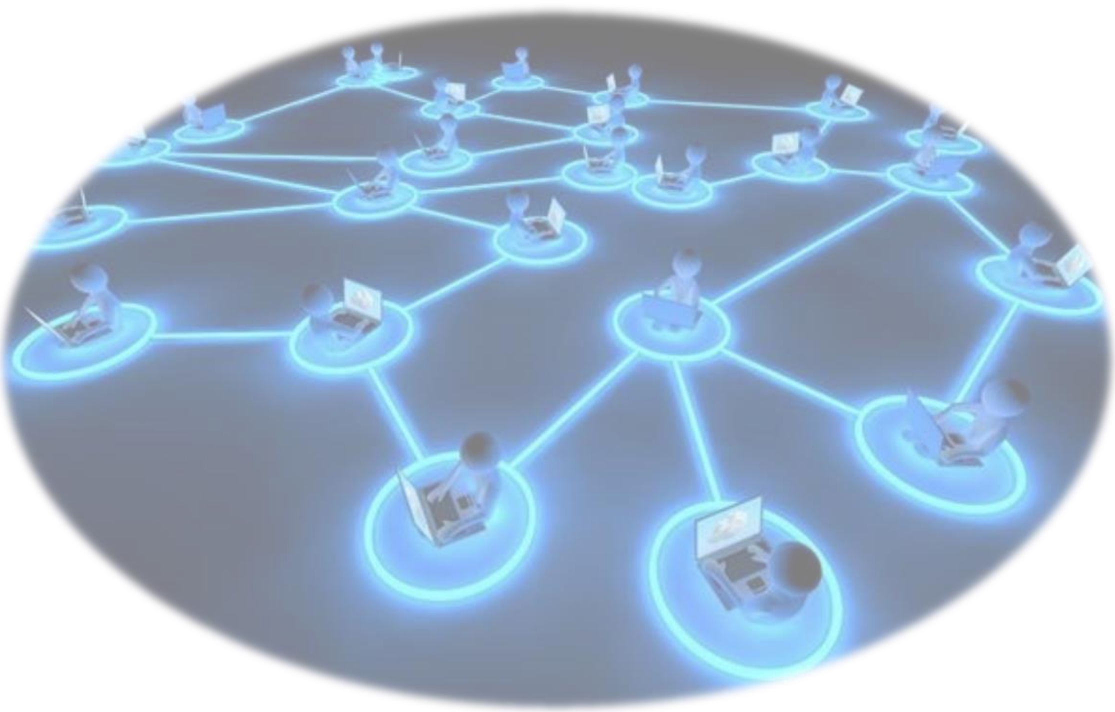
The network design project undertaken by PTHK for the CLIENT involved applying the first four phases of the 5 Phase Network Design methodology to evaluate the current network infrastructure and provide necessary suggestions to improve productivity, overall customer, and company satisfaction.

Executive Overview (continued):

During the first phase, PTHK conducted a walkthrough and network user survey to determine the needs of the users, business, and applications. Subsequently, the Requirements Gathering phase was presented on March 2, 2023, outlining the requirements gathered from the survey.

In the second and third phases, PTHK analyzed the network traffic and created a logical diagram of the current and updated network infrastructure. The Traffic Analysis & Logical Design phase was presented on March 30, 2023, providing an overview of the analysis and design of the network infrastructure.

At present, PTHK has completed the fourth phase of the project, which involved creating physical diagrams of the current and updated network infrastructure. These physical diagrams depict the proposed network infrastructure design and are included in this document.



Project Review by Phase – Requirements Gathering:

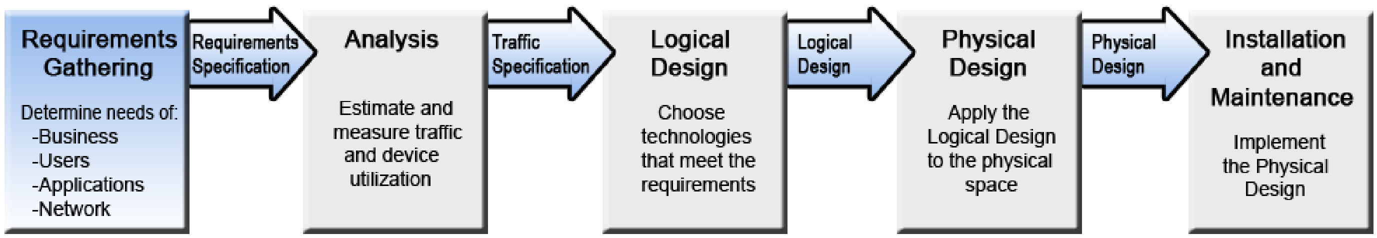


Figure 3

As previously mentioned, the requirement gathering phase is the first step in the network design process undertaken by PTHK. To complete the Requirement Collection phase, the following techniques were used to contact users and investigate current needs for CLIENT Solutions:

- A **walkthrough** was held at CLIENT Solutions on 20th January 2023. Meeting with *Mr. X*, an IT Associate, focused on a general overview of the current network infrastructure at the Client. Local Area Network connections, cloud and servers have been discussed in brief.
- The **user survey** was conducted at **CLIENT** using web-forms by **PTHK.** *Mr. X* (IT Associate), *Mr. X* (IT Specialist), *Mr. X* (Business Developer), and a few other members from **CLIENT** participated in the survey. The User Survey was divided into 4 sections personal computer, ergonomics, network performance, and security. As a part of each individual section many questions were listed to gather information about user's computer specifications, computer usage, user comfort at the work area, **CLIENT's** network specifications and security. To get more knowledge about the survey please search for Appendix A, B, and C at the end of this document.
- During the **Requirement Gathering Phase,** following constraints were illustrated:
  - Project Completion Date: April 31st, 2023.

Following information gathered while completing the **Requirement Gathering Phase:**

- Internet speed for wired and wireless connections is good and it has 50 MB/50 MB powered by MNSI through fiber cabling going through the building.
- **CLIENT** is using AWS for all the services including emailing, Active Directory, and storage servers. Also, auto backup occurs 3 times a day for entire Microsoft 365 suits and Active Directory (SharePoint, mailbox, user profile included) by Datto.
- **CLIENT**'s network specifications, details, maintenance reports and requirements are stored in Excel for keeping track of the network.
- Cisco ASA 5516-X firepower, firewall is being used for cyber security rules implementation on exchange pertaining to phishing and impersonation scams.
- For secure remote access to Avigilion Server (Protégé Server), VPNs are implemented.

Project Review by Phase – Requirement Gathering (continued):

- According to equipment usage on **CLIENT's** network, there are around 70 laptops throughout the plant and office for day-to-day use by employees, 3 brother office printers, 2 label printers, and employees' personal devices.
- Equipment specification for the one provided to employees are as follows: Dell Inspiron 5000 powered on Windows 10/11 Home, Logitech wireless mouse & keyboard, Dell S2421NXs @ 60hz monitors, and Bell work cellphones. Because of these specifications computers at **CLIENT** are capable of meeting work requirements.
- IT Loft Rack is equipped with Catalyst 9300 48 Port POE Network Essential and Cisco Catalyst 9200L 48 Port POE Network Essential switches, Cisco ASA 5516-X firewall, and UPS and 715 Power Supply for emergencies.
- As a part of ergonomics, **CLIENT** makes sure the all the employees are well suited with equipment to remain comfortable during prolonged sitting periods, also encourages to have some talking walking breaks.

The following graphs present the results obtained from the network user survey conducted by PTHK.

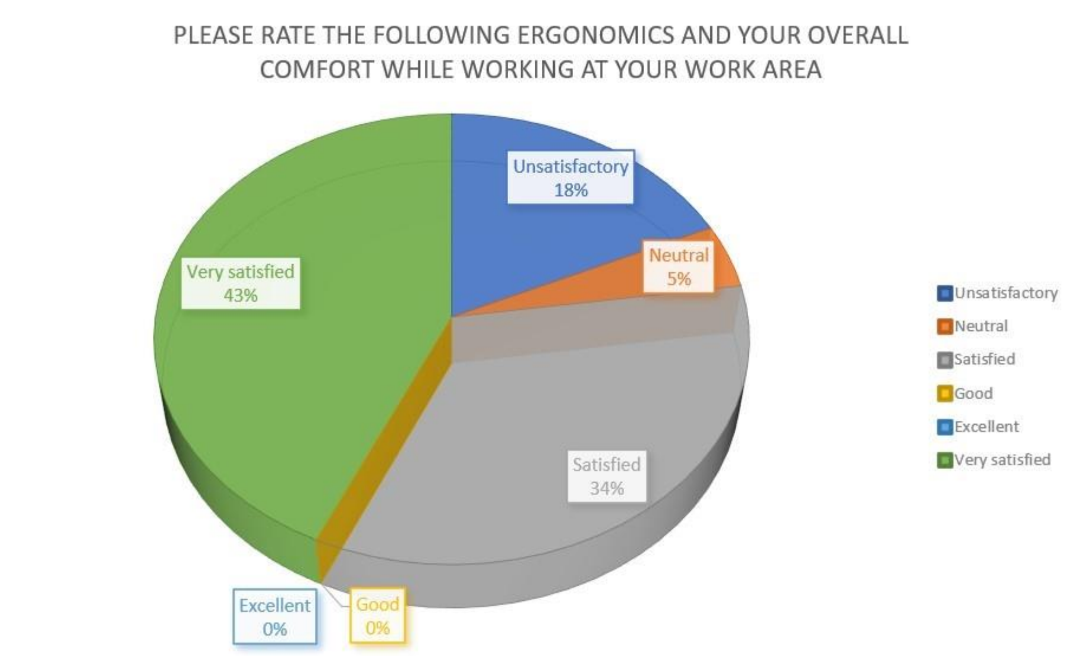


Figure 4

The survey results for the ergonomic workspace were compiled in the graph below by CLIENT Solution. Many workers are content with the ergonomics and general comfort of their workspace.

Project Review by Phase – Requirements Gathering (continued):

The following graph presents the results of network performance.

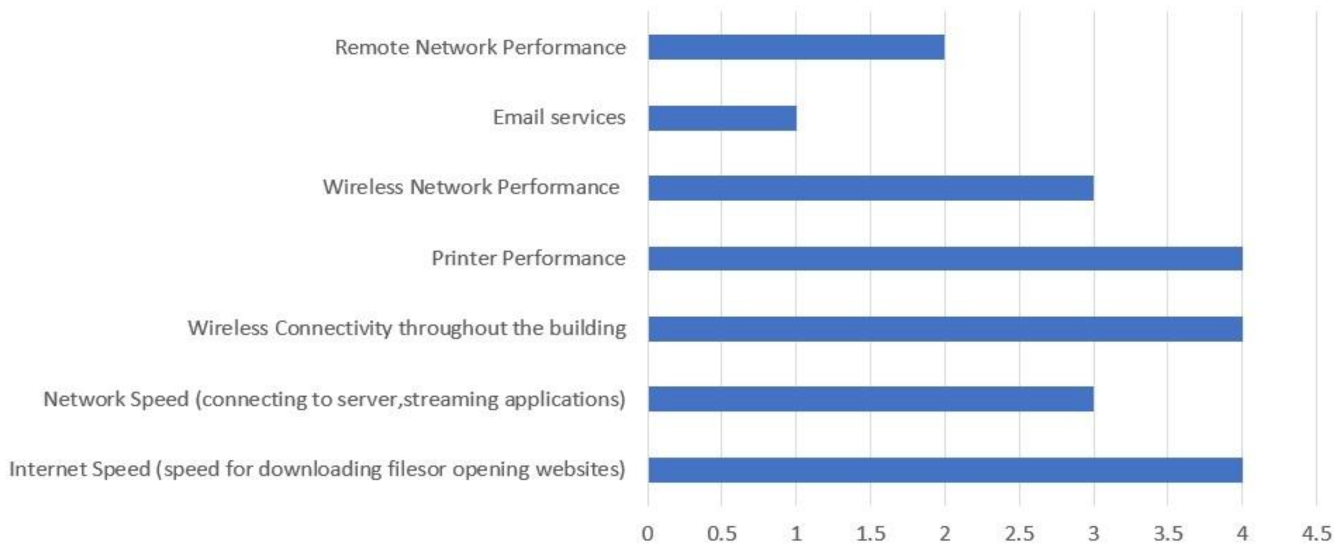


Figure 5

The results of the survey taken for the CLIENT Network were shown in the graph above. The graph demonstrates that there aren't many network-related problems for employees, likewise, wireless network connectivity, its speed, etc. That is why there is a need for wireless network enhancement.

Project Review by Phase – Requirements Gathering (continued):

The following figure illustrates the opinions of employees about the workstation's performance at CLIENT.

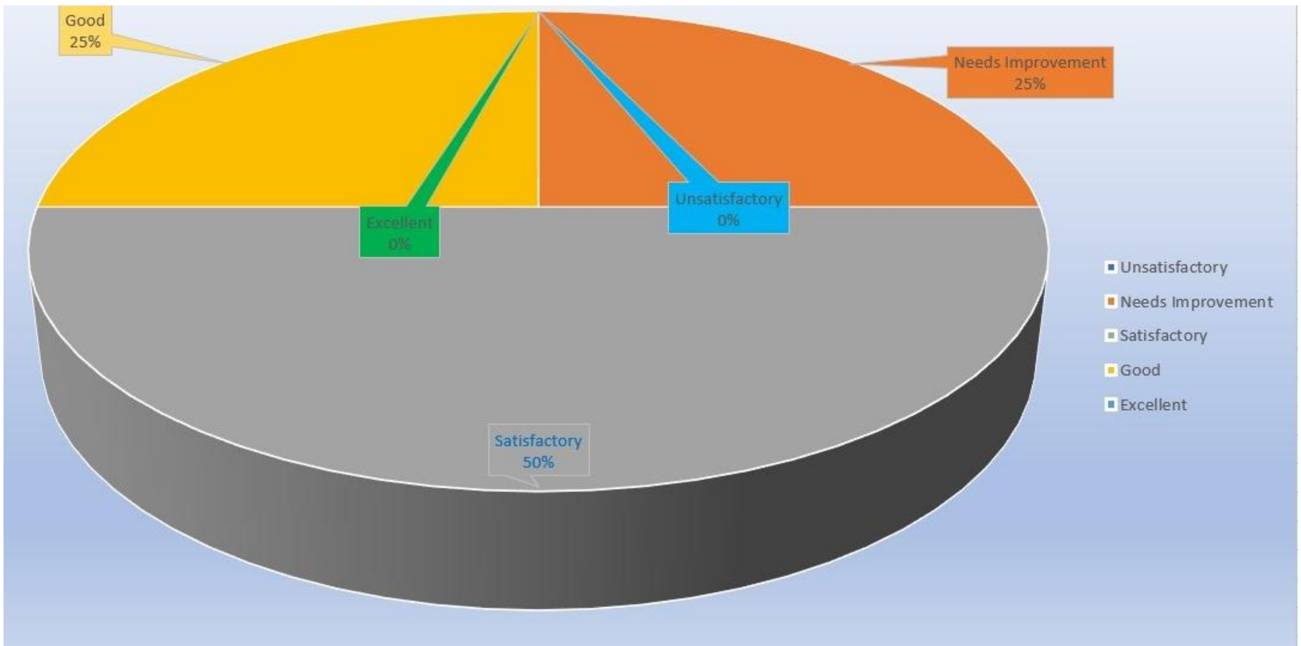


Figure 6

Figure 5 displays the results of a survey on the overall performance of employees' workstation PC. The graph indicates that 50% of the workforce is satisfied with PC performance, but there are areas where the PC workstation could be improved.

Project Review by Phase – Requirements Gathering (continued):

Here are some of the Priorities Recommendations list that **PTHK** discovered during the Requirement Gathering Phase:

1. **Improve Network Backbone – To Gain Wireless Network Speed**
   During the initial meeting with Mr. X, it was revealed that the network speed throughout the plant is currently limited to 50 Mbps/50 Mbps. This bandwidth has proven inadequate for streaming certain applications utilized during CLIENT operations. Notably, during periods of high employee activity, such as when accessing cloud resources or servers, as well as during normal operations when large files are being loaded from the cloud, individuals may experience extended wait times.

2. **Improve Employee Workstation Quality – To Use Latest Software and Reduce Loading Time In A Network**
   As per the results of the User Survey, it has been identified that additional RAM and new components are required to improve the performance of the workstations. Presently, the computers take approximately 40 seconds to display the logon prompt and an additional 30 seconds to become operational from the home screen. This delay is primarily due to the heavy usage of the Office 365 suite and Epicor software by the employees, which utilizes a significant portion of the CPU and RAM of the workstations. Furthermore, certain software applications necessitate the latest version of the operating system for optimal performance.

3. **Improve IT Rack Quality – To Manage Network Components Easily with Better Cooling System**
   During the initial site walkthrough, PTHK ascertained that the IT Rack sponsored by the site is insuffiCliently sized to accommodate additional network components. Moreover, the IT Rack lacks a proper cooling system to effectively regulate temperature and maintain the network components at an optimal operating level.

4. **Improve Ergonomics – To Provide Comfortable Workspace to Employees**
   During the meeting with Mr. X held while distributing the User Survey, it was discovered that the staff desks currently in use are not adjustable and lack suffiClient height to allow for comfortable working conditions. Additionally, the limited length of the tables presents a challenge for employees, who are required to position their workstations in close proximity, which may result in eye strain and discomfort.

Project Review by Phase – Traffic Analysis:

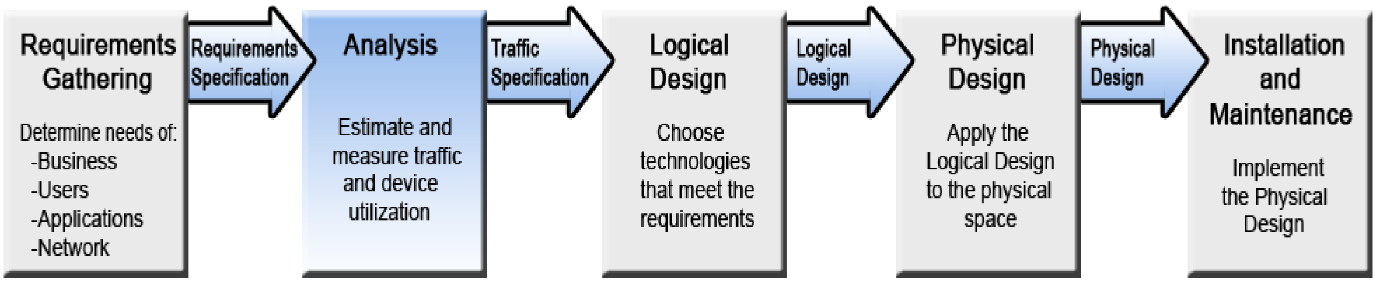


Figure 7

In this phase, the existing network infrastructure is evaluated and the requirements for the new network are further refined. To complete the Traffic Analysis phase, the following techniques were used to contact users and investigate current needs for CLIENT Solutions:

- A **meeting** was held with Mr. X, an IT Associate at **CLIENT** on the 8th of March. The meeting was related to traffic capturing with connecting span port to laptop's ethernet. The laptop was made available with required configurations including the bat file and network port disabled. The captured process was made successful using certain kinds of applications like Wireshark and Task Scheduler. 900 Megabytes of data was captured for one day with a duration of 7 minutes for every half-an-hour interval. The total number of packets captured in this period is 6436267.
- **Wi-Fi Mapping** process was placed on March 17 at **CLIENT**. Mr. X guided throughout the visit of **PTHK** to measure Wi-Fi signal strengths, connectivity, and speed. Additionally, **PTHK** covered general Q&A to gather more information about CLIENT's network requirement.

Following key points were discovered while completing the **Traffic Analysis Phase**:

- Traffic estimates for E-Mail, FTP Server, and Security Cameras are as follows: 0.144 Mbps, 12 Mbps, and 4.8 Mbps as shown in the section below with calculation.
- Internet speed for wired and wireless connections is not good enough, which has 50 Mb/50 Mb powered by MNSI through fiber cabling going through the building. This is measured using Wi-Fi mapping.
- Dell Inspiron 5000 powered on Windows 10 Home which is not good enough to use some of the additional features provided by Windows 10 Pro edition. Specifically, on Windows 10 Pro they could conveniently use virtual machines. Also, RAM is an issue with workstations as workstations only have 8 GBs of RAM and majorly, employees use Office 365 suits, and Epicor together which leads to more RAM and CPU utilization.
- As for the transmission packets, network traffic contains majorly multicast addresses than broadcast.
- According to the frame distribution this result has been described that packets length from 1280 to 2559 covers almost 45% of the network traffic.

Project Review by Phase – Traffic Analysis (continued):

Illustrated below, Table 1 represents the estimated total requirements for the backbone traffic.

Application	% Distribution per Segment	# of Simultaneous Sessions	Average Transaction Size	Estimated Total Capacity Required
E-Mail	**100%**	0.09	200KB	0.144Mbps
FTP Server	**100%**	1	1500KB	12Mbps
Security Cameras	**100%**	0.75	800KB	4.8Mbps
			Totals	16.944Mbps

Table 1

Step 1

Determine the Average Transaction Size and Number of Simultaneous Sessions per second.
ATS for Office 365 E-mail = 200KB Sessions per Second = 0.09

Step 2

Multiply these two values together to determine the estimated bandwidth requirements per second.
200KB x 0.09/second = 18KB/second

Step 3

Convert KB (Kilobytes)/second to Kb (Kilobits/second)
18KB/s x 8 = 144Kb/sec

Step 4

Convert Kb per second into Mb per second, (division by 1000 is suffiClient for estimation purposes).
144Kbps ÷ 1000 = 0.144Mbps

Step 5

The approximate bandwidth requirement for a network is the total estimated capacity of all applications.
0.144Mbps + 12 Mbps + 4.8 Mbps = 16.944 Mbps

Project Review by Phase – Traffic Analysis (continued):

Following graph depicts the detailed information about Frame size Distribution:

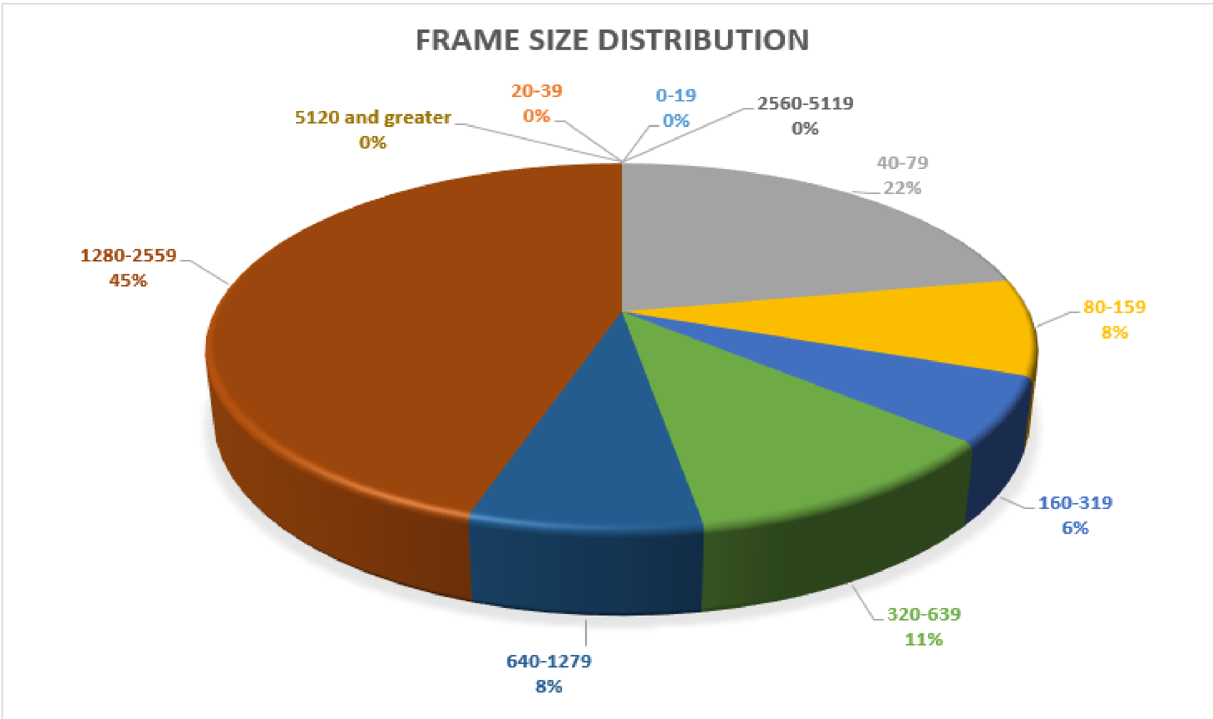


Figure 8

This work presents a graph that displays the distribution of packet sizes in the network. The graph shows that there were no packets transmitted within certain size ranges, while other size ranges had varying percentages of transmitted packets. The largest percentage of transmitted packets fell between 1280 and 2559 bytes in size, while the second largest percentage fell between 40 and 79 bytes in size. This information can be helpful for evaluating network performance and identifying any potential issues.

Project Review by Phase – Traffic Analysis (continued)**:**

Following graph describe the detailed information about broadcast and multicast traffic:

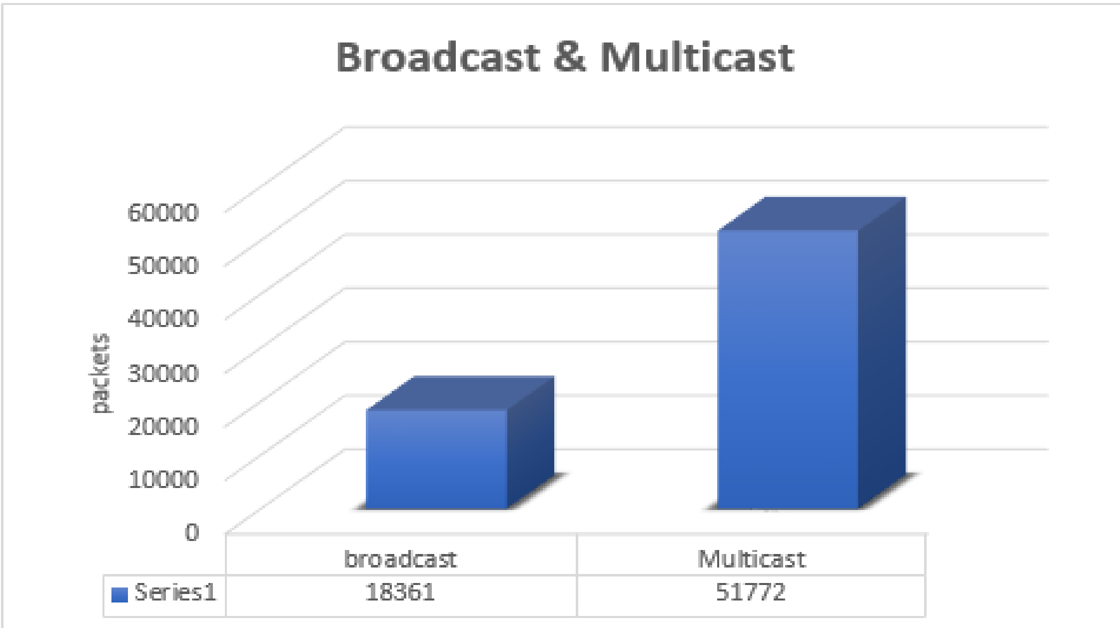


Figure 9

In this graph above shows the broadcast and multicast, Broadcast transmission sends data from one source to all devices on a network, while Multicast sends data from one source to a select group of devices on the network. Broadcast is used to send the same data to all devices, while multicast is used to send the same data to a specific group of devices. Multicast is commonly used for streaming media, video conferencing, and other applications that require simultaneous transmission of data to multiple devices. All devices on the network received the same information because 18,361 broadcast packets carrying the same data were sent. 51772 multicast packets were broadcast to a chosen set of network nodes.

Project Review by Phase – Traffic Analysis (continued)**:**

Following diagram describe the detailed information about CLIENT’s network usage:

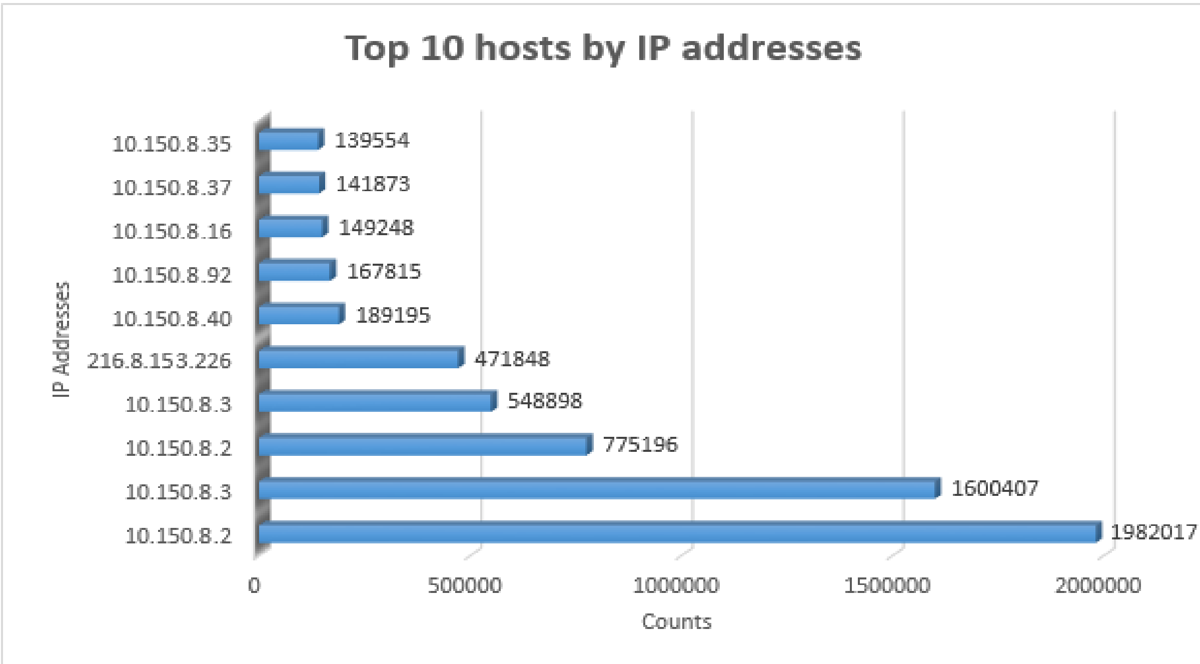


Figure 10

The graph displays the top 10 hosts based on their IP addresses and the corresponding number of frames sent or received by each host. Analyzing this graph can provide insights into network traffic patterns and potential security threats. Upon inspection, it becomes evident that the host with the IP address 10.150.8.2 is responsible for transmitting or receiving the most data across the network.

Project Review by Phase – Traffic Analysis (continued)**:**

Following picture demonstrates the CLIENT's wireless network mapping:

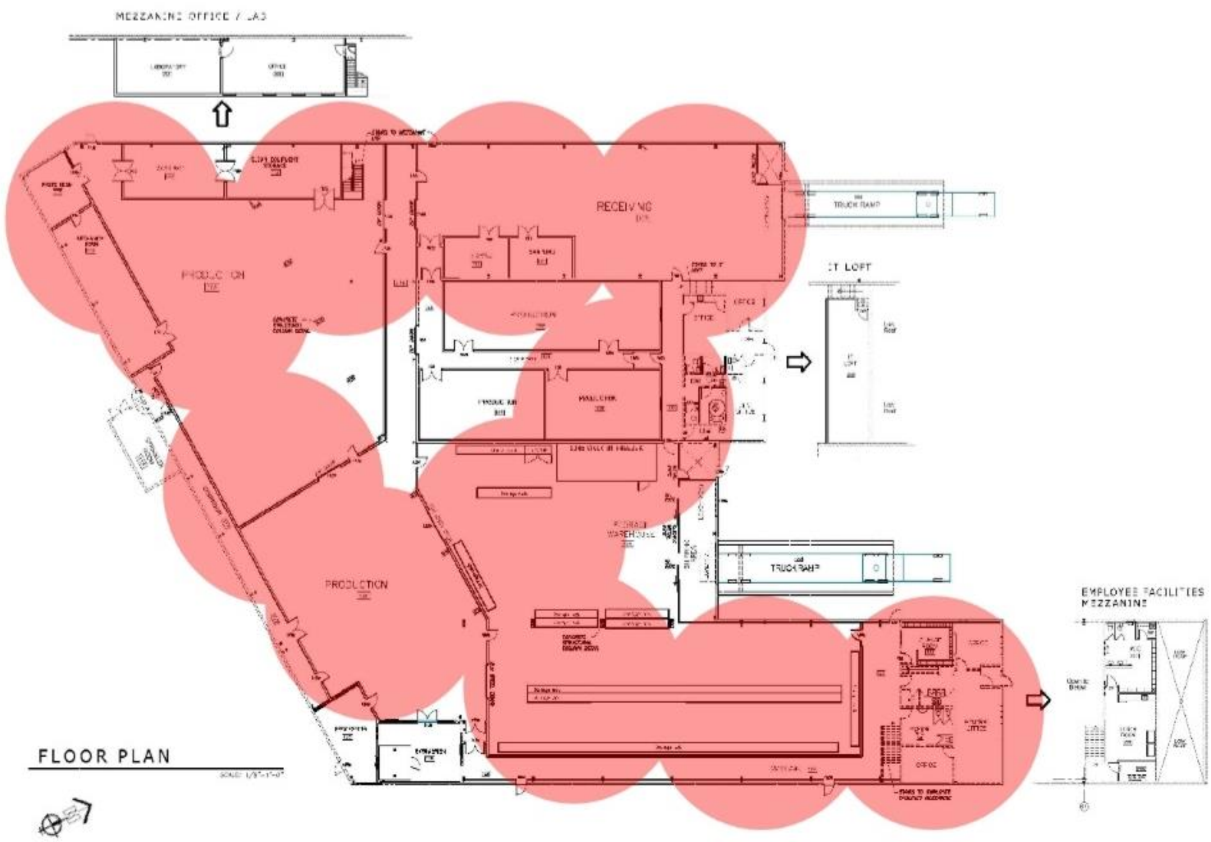


Figure 11

Based on the diagram presented above, it appears that the wireless network speed, frequency, and usage of access points (APs) are not optimal for the CLIENT network. Despite the presence of 30 APs, insuffiClient network speed and improper AP placement are contributing factors to this issue.

Project Review by Phase – Logical Design:

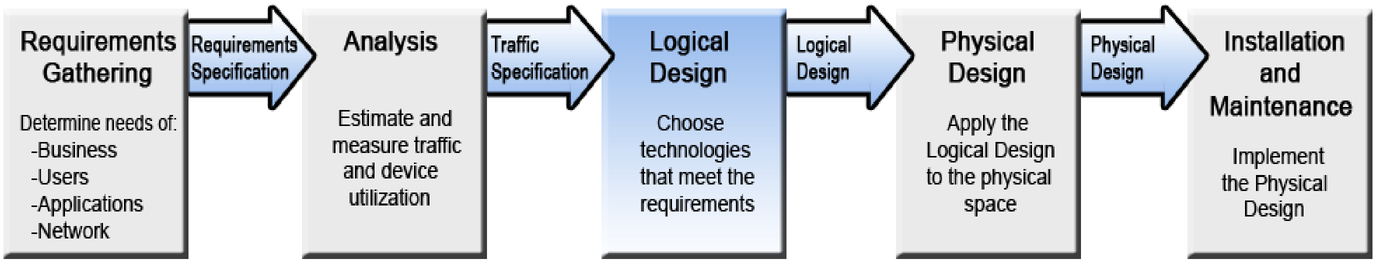
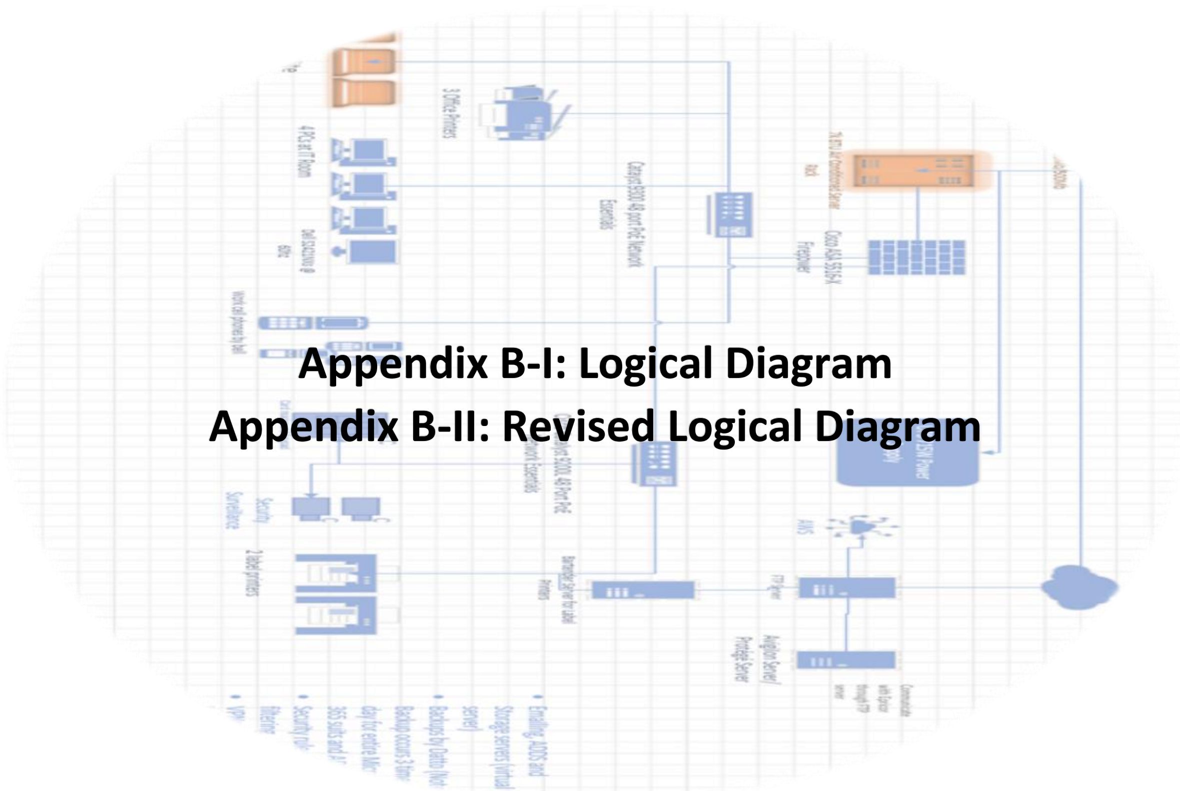


Figure 12

As mentioned earlier in this phase, the detailed network design is created, including the selection of hardware and software components and the creation of network diagrams.



Appendix B-I: Logical Diagram

Appendix B-II: Revised Logical Diagram

Project Review by Phase – Logical Design (continued):

Based on the findings from traffic analysis and previous meetings, we have compiled a list of recommendations along with a summary of the current network components.

1. **Upgrade Wi-Fi Router – To Increase Wireless Network Speed**
   Current Router: MNSi Fibe Internet with 50/50 Mbps

Option 1A

MNSi for Business

500/500 Mbps Fibe Internet

Unlimited monthly Usage, Static IP address included, and 3-Year Contract.

Option 1B

Gigabit Business Fibe

940/940 Mbps

Unlimited monthly Usages, Bell Protection, 3-Year term plan.

Product	Quantity	Price	Total Price
MNSi for Business	1	$125.00	$125.00
Gigabit Business Fibe	1	$94.95*	$94.95

Based on the current connectivity situation and potential fluctuations in internet pricing during the contract, it is recommended to consider MNsi as a superior option compared to Bell.

Project Review by Phase – Logical Design (continued):

2. Upgrade Workstation – To Increase Overall Performance and Functionality

Current Workstation: Dell Inspiron 5000
Processor: Intel Core i5 8th Gen 8250U 1.6Ghz
RAM: 8GB
Storage: 256GB HDD
Graphics: AMD Radeon 530

Option 2A

HP Elite Dragonfly Notebook

CPU: Intel Core i7 8th Gen
OS: Windows 10 Pro
RAM: 16 GB LPDDR3
Storage: 256 GB SSD

Option 2B

Lenovo ThinkPad X1 Carbon

CPU: Intel Core i7 7th Gen
OS: Windows 10 Pro
RAM: 16 GB
Storage: 1 TB SSD

Product	Quantity	Price	Total Price
HP Elite Dragonfly Notebook	70	$3053.38	$213,736.60
Lenovo ThinkPad X1 Carbon	70	$4057.78	$284,044.60

The HP Elite Dragonfly is a recommended option to consider, as it offers a lower cost compared to other laptops. Additionally, with work primarily stored on the cloud, a storage capacity of 256 GB SSD should suffice.

Project Review by Phase – Logical Design (continued):

3. Upgrade Server Rack – To Improve Cooling and System Stability

Option 3A

7k BTU Air-Conditioned Server Rack

Rack Units: 42
Cooling Capacity: 7000 BTU (2.0 kW)
Noise Level: 57 dbA
Door: Vented Door

Option 3B

Rb-Av42 42U 24D Audio-Visual Cabinet

Rack Units: 42
Cooling Capacity: NA
Door: Glass Door

Product	Quantity	Price	Total Price
7k BTU Air Conditioned Server Rack	1	$3857.58	$3857.58
Rb-Av42 42U 24D Audio-Visual Cabinet	1	$2142.86	$2142.86

The recommended option to consider would be the 7k BTU Air-Conditioned Server Rack due to its built-in air-cooling system and lower noise levels compared to other alternatives.

Project Review by Phase – Logical Design (continued):

4. Upgrade Staff Desk – To Improve Work Area Ergonomics

Option 4A

Altamae 48" Adjustable Height Standing Desk

Additional 5-year Protection Plan for $48.99
Width: 48"
Warranty: 3 Years
Weight Capacity: 180 lb
Material: Metal

Option 4B

RODULF Desk sit/stand

Length: 56"
Width: 32.5"
Weight Capacity: 154 lb
Color Option: Gray/White

Product	Quantity	Price	Total Price
Altamae 48" Adjustable Height Standing Desk	50	$296.99	$14,849.50
RODULF Desk sit/stand	50	$429.00	$21,450.00

The Altamae 48" Adjustable Height Standing Desk presents a compelling option, given its comparative affordability in relation to alternative products. Moreover, it includes the added benefit of a 3-year warranty, further enhancing its value proposition.

Project Review by Phase – Physical Design:

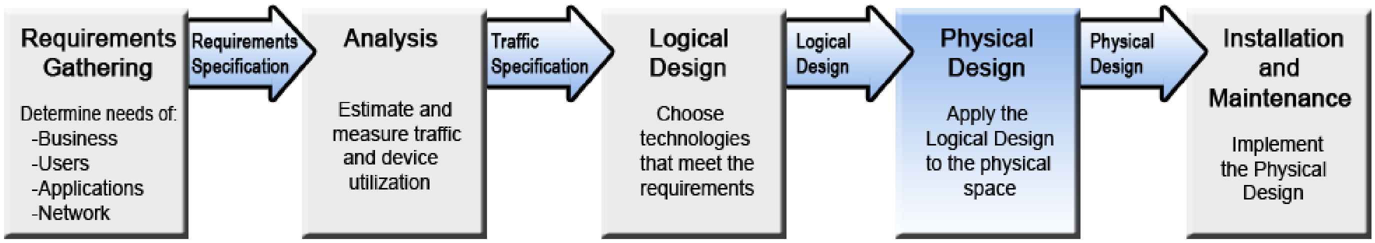


Figure 13

Now, in this phase, the network design is put into action through installation, configuration, and testing of the network components.

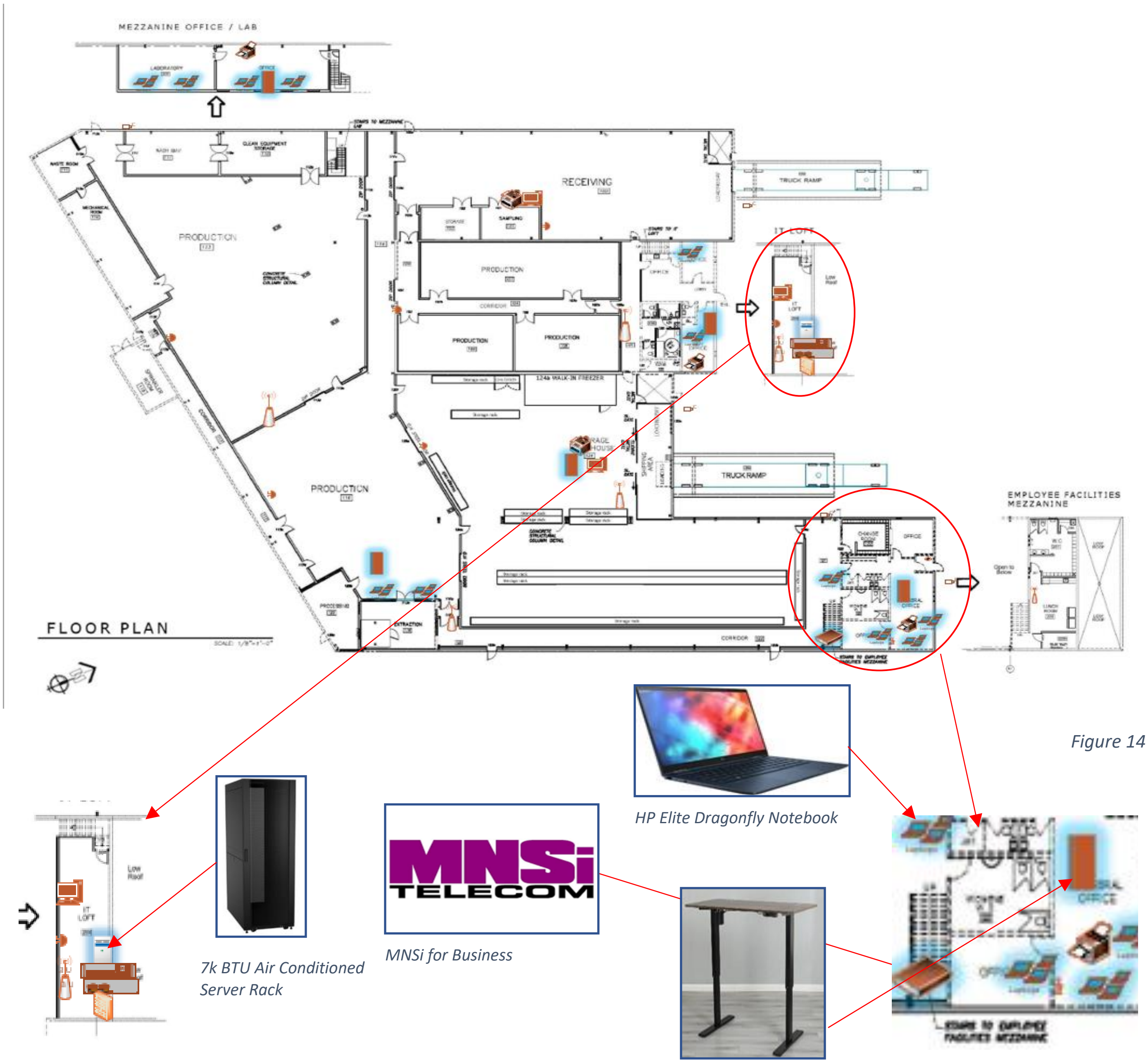
Here is the asset list that outlines the new equipment that must be procured, existing equipment, and equipment requiring disposal.

	Asset List - CLIENT – April 2023				
Device	**Model**	**Quantity**	**Location**	**Serial #**	**Comments**
New Items – To Be Purchased					
Routerr	MNSi for Business	1	Hallway		
Laptops	HP Elite Dragonfly Notebook	70	Plant Wide		
IT RACK	7k BTU Air-Conditioned Server Rack	1	Server Room		
Staff Desk	Altamae 48” Adjustable Height Standing Desk	50	Plant Wide		
Existing Technology – To Be Re-Used					
Switches	Cisco Catalyst 9200L 48 Port PoE	2	Sever Room		
Firewall	Cisco ASA 5516-X	1	Server Room		
UPS	715W Power supply	1	Server Room		
Monitors	Dell S2421NXs @ 60hz	20	Plant Wide		
Keyboard	Logitech	20	Plant Wide		
Mouse	Logitech	20	Plant Wide		
Printers	Label Printers	2	Production Area		
	Office Printers	3	Plant Wide		
Cat 5e Cable	Various Manufacturers	All	Plant Wide		
Fiber Cable	Various Manufacturers	All	Plant Wide		
Existing Items – To Be Discarded					
Laptops	Dell Inspiron 5000	70	Plant Wide		Return on lease

Table 2

Project Review by Phase – Physical Design (continued):

The following Revised Physical Diagram depicts the changes that are made according to the recommendation list from Traffic Analysis and Logical Design Phase illustrations. The comment for this diagram is demonstrated on the next page.



HP Elite Dragonfly Notebook

Figure 14

7k BTU Air Conditioned Server Rack

MNSi for Business

Altamae 48” Adjustable Height Standing Desk

Project Review by Phase – Physical Design (continued):

Notes & Comments:

The Revised Physical Diagram, displayed on the preceding page, includes various modifications to the site's infrastructure. The diagram features security surveillance surrounding the site, an installation of approximately 70 new laptops, a new staff desk in the office area, and a new modem situated in the corner of the primary hallway. Additionally, roughly 30 access points, printers, and PCs have been installed across the site to support production and receiving area activities, along with smoke and fire alarms for added safety measures. Notably, the IT room has undergone changes with the installation of a new server rack, accompanied by pre-existing network components such as switches, firewalls, and UPS power supplies.

The following Gant Chart gives an idea about the project milestones and the process that has been done till date.

Network Design Process	**Thu 1/12/23**	**Fri 4/28/23**	**77 days**
Requirement Gathering Phase	**Thu 1/12/23**	**Thu 3/2/23**	**36 days**
Sponsor Confirmation	Thu 1/12/23	Thu 1/12/23	0 days
First on site Visit	Fri 1/20/23	Fri 1/20/23	0 days
Feasibility Presentation	Thu 1/26/23	Thu 1/26/23	0 days
Survey Construction	Mon 2/6/23	Mon 2/20/23	11 days
Business & Learning Goals Approval	Sun 2/19/23	Sun 2/19/23	0 days
Survey Approval & Administration	Wed 2/22/23	Wed 2/22/23	0 days
R. G. Specification Document Finalized	Wed 3/1/23	Wed 3/1/23	0 days
R. G. Presentation	Thu 3/2/23	Thu 3/2/23	0 days
Traffic Analysis & Logical Design Phase	**Wed 3/1/23**	**Thu 3/30/23**	**22 days**
Sniffer Booking	Wed 3/8/23	Thu 3/9/23	2 days
Data Capture Completed	Thu 3/9/23	Thu 3/9/23	0 days
Additional System Measurements	Fri 3/17/23	Fri 3/17/23	1 day
Wi-Fi Heat Mapping	Fri 3/17/23	Fri 3/17/23	1 day
Logical Diagrams & Solutions	Wed 3/22/23	Tue 3/28/23	5 days
TA & LD Specification Document Finalized	Wed 3/29/23	Wed 3/29/23	1 day
TA & LD Presentation	Thu 3/30/23	Thu 3/30/23	0 days
Physical Design Phase	**Wed 4/5/23**	**Fri 4/28/23**	**18 days**
Final Document Review	Fri 4/21/23	Fri 4/21/23	0 days
Physical Design Diagrams	Sat 4/22/23	Sat 4/22/23	1 day
Final Cost Summary & Gant Charts	Sun 4/23/23	Sun 4/23/23	1 day
Final Document Completed	Mon 4/24/23	Mon 4/24/23	1 day
Sponsor Thank you	Fri 4/28/23	Fri 4/28/23	1 day
Final Presentation	Fri 4/28/23	Fri 4/28/23	0 days

Figure 15

Project Review by Phase – Physical Design (continued):

The Gantt chart presented after provides insight into the hours of work associated with the network changes.

1 Implementation Phase	**64 hrs**	**14 days**
1.1 Purchase Equipment	**4 hrs**	**4 days**
1.1.1 Purchase MNSi for Business	1 hr	1 day
1.1.2 Purchase HP Elite Dragonfly Notebook	1 hr	1 day
1.1.3 Purchase 7k BTU Air-Conditioned Server Rack	1 hr	1 day
1.1.4 Purchase Altamae 48" Adjustable Height Standing Desk	1 hr	1 day
1.2 Server Room Management	**18 hrs**	**3 days**
1.2.1 Remove Existing Server Rack	4 hrs	1 day
1.2.2 Install Server Rack	4 hrs	1 day
1.2.3 Install Network Components	5 hrs	1 day
1.2.4 Install Cooling systems	5 hrs	1 day
1.2.5 Server Rack Installation Complete	0 hrs	0 days
1.3 Cabling Infrastructure	**12 hrs**	**2 days**
1.3.1 Install Cable	8 hrs	1 day
1.3.2 Review Path Panel and End Devices	4 hrs	1 day
1.3.3 Cable Installation Complete	0 hrs	0 days
1.4 Wireless Network Management	**3 hrs**	**1 day**
1.4.1 Install MNSi for Busines	2 hrs	1 day
1.4.2 Test Wireless Network Speed	1 hr	1 day
1.4.3 Wireless Network Setup Complete	0 hrs	0 days
1.5 Wrokstation Management	**16 hrs**	**2 days**
1.5.1 Install Workstations	8 hrs	1 day
1.5.2 Test Workstation Perfomance	8 hrs	1 day
1.5.3 Workstation Installation Complete	0 hrs	0 days
1.6 Workarea Ergonomics	**11 hrs**	**2 days**
1.6.1 Remove Existing Staff Desk	5 hrs	1 day
1.6.2 Install Altamae 48" Adjustable Height Standing Desk	6 hrs	1 day
1.6.3 Altamae 48" Adjustable Height Standing Desk Installation Complete	0 hrs	0 days

Figure 16

Project Review by Phase – Physical Design (continued):

The following tables offer a broad understanding of the anticipated expenses for equipment procurement and the necessary labor costs for installation. Note: Labor Cost is $100 per hour for this job.

Equipment Cost Analysis

Option	Product	Price	Units	Subtotal
1A	MNSi for Business	$125.00	1	$125.00
2A	HP Elite Dragonfly Notebook	$3053.38	70	$213,736.60
3A	7k BTU Air-Conditioned Server Rack	$3857.58	1	$3,857.58
4A	Altamae 48" Adjustable Height Standing Desk	$296.99	50	$14,849.50
		Sub Total		$2,32,568.68
		HST 13%		$30,233.93
		Total		$2,62,802.61

Table 3

Total Cost Summary

Item (Including Taxes)	Price
Equipment Cost	$2,62,802.61
Labor Cost	$7,232.00
Grand Total	$2,70,034.61

Table 5

Labor Cost Analysis

Option	Hours	Price
1A	4	$400.00
2A	17	$1,700.00
3A	31	$3,100.00
4A	12	$1,200.00
	Sub Total	$6,400.00
	HST 13%	$832.00
	Total	$7,232.00

Table 4

Approval Section:

PTHK has completed the first four Phases of the network design process. During the first three Phases the needs of business, users, applications, and network were determined; estimated and measured traffic and device utilization; chose the technologies that meets the requirements and summarized in this specification Document.

As of now, Network Design Process is Phase 4, the Physical Design Phase has been completed. In this Phase **CLIENT** has considered to apply logical design in the physical space.

By signing this document, below, **CLIENT**, agrees to the findings presented within, and authorizes **PTHK** to proceed with Phase 5, The Installation and Maintenance Phase.

☐ Approved

Not Approved ☐

CLIENT
Oldcastle, ON N0R 1L0

PTHK

Mr. X IT Associate	/ / Date	Kishan Panchal KP93@MYSCC.CA	/ / Date
		Siem Telkleweini ST62@MYSCC.CA	/ / Date
		Markus Hanna MH126@MYSCC.CA	/ / Date
		Harjas Kaur HK221@MYSCC.CA	/ / Date

Comments:

General:
<u>Documentation</u> <u>Slides</u>
<u>Presentation</u>

Professor	**Professor**	**Professor**	**Professor**
Professor	**Professor**	**Professor**	**Professor**
Professor	**Professor**	**Professor**	

Corporate Sponsor Thank You:

April 21, 2023

Mr. X (he/him) | IT Associate

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Dear **Mr. X**,

The **PTHK** team wanted to reach out to you and express our sincerest appreciation for your sponsorship of our project. It has been almost 4 months since you joined us as our sponsor through CLIENT, and we could not be more grateful for your support.

Your contribution has been instrumental in helping us achieve our learning goals, and we wanted to take this opportunity to thank you for partnering with us in this endeavor. We feel privileged to have had the opportunity to work with CLIENT, an organization that is dedicated to creating a positive impact in our community.

Once again, thank you for your generous support.

Best Regards,

____________________	____________________	____________________	____________________
Kishan Panchal	Siem Tekleweini	Harjas Kaur	Markus Hanna